Wave Speed in a fluid

$$\text{wave speed} = \left[\frac{\text{restoring force}}{\text{inertia}}\right]^{1/2}$$

for mechanical waves

- For a bulk fluid, the restoring force is related to how difficult it is to compress the fluid. → Bulk modulus, B

$$B = -V_0 \frac{\Delta P}{\Delta V}$$

$\Delta V \sim$ volumetric Δ

$P \sim$ pressure

$V_0 \sim$ initial volume

Check units for B?

- The inertia/massiveness of the bar fluid is described by $\rho = m/\Delta V$
- Speed of longitudinal wave:

$$v = \sqrt{\frac{B}{\rho}}$$

$$B = -V_0 \frac{\Delta P}{\Delta V}$$

(−) means positive change in P translates to negative change in volume.

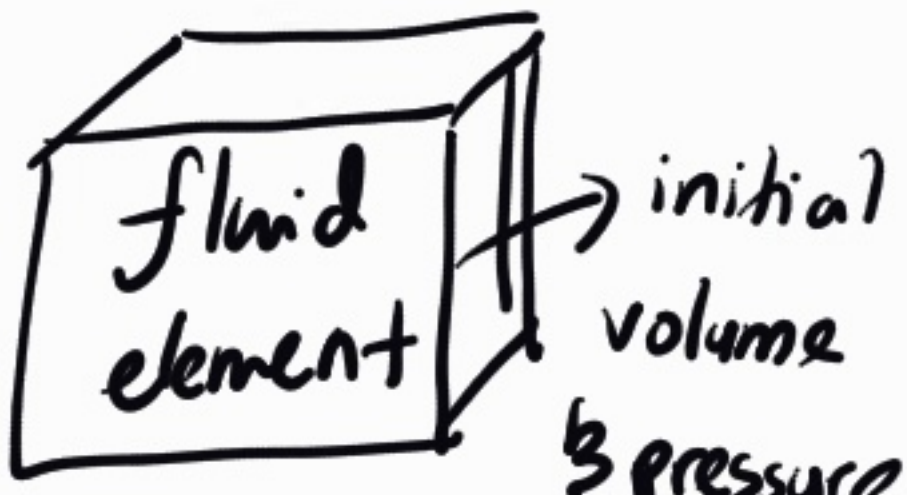
recall for a string

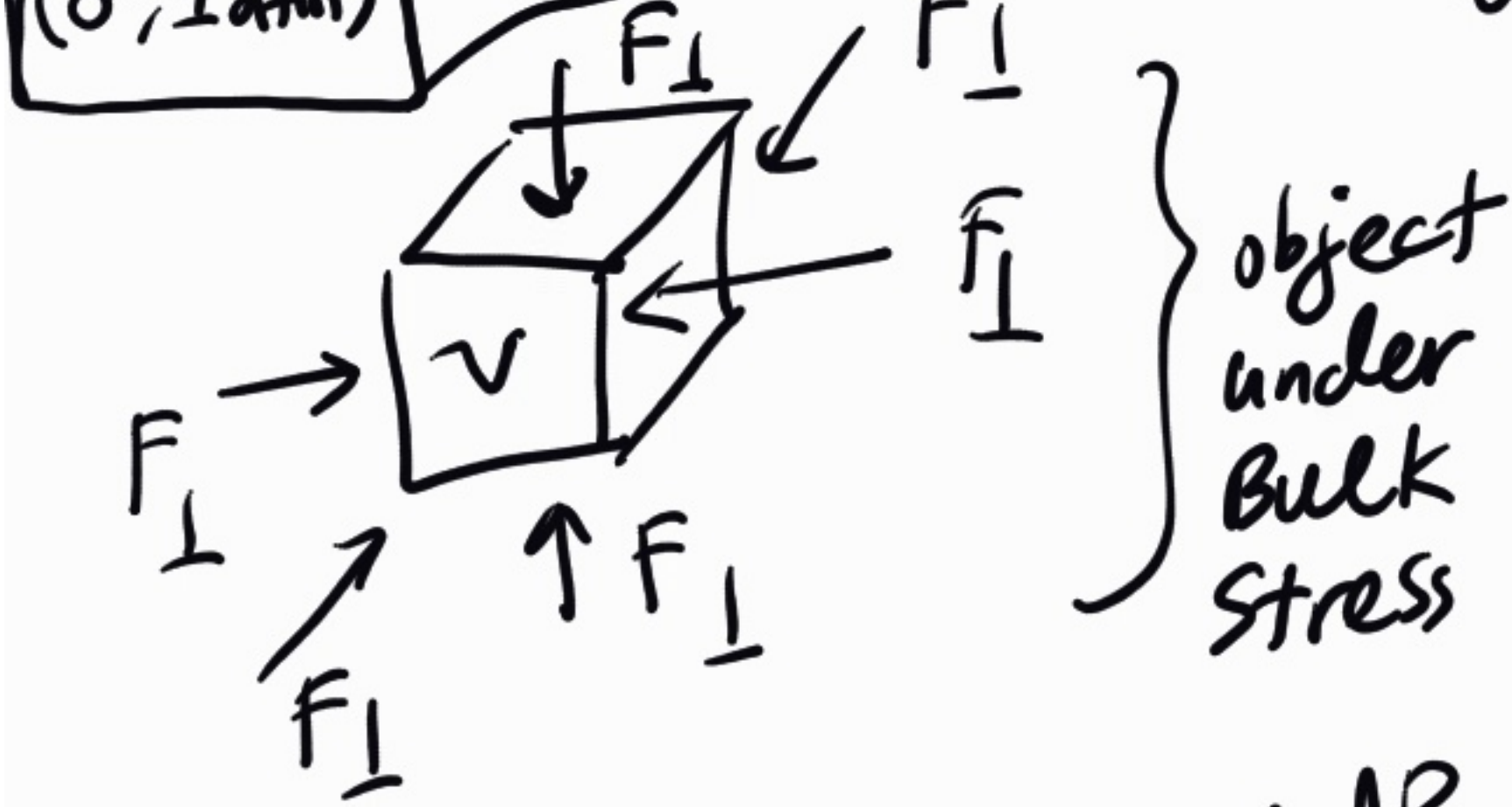
$$v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{\text{tension}}{\text{linear density}}}$$

Wave Speed in a Fluid.

$$\Delta P = -B \Delta V / V_0$$

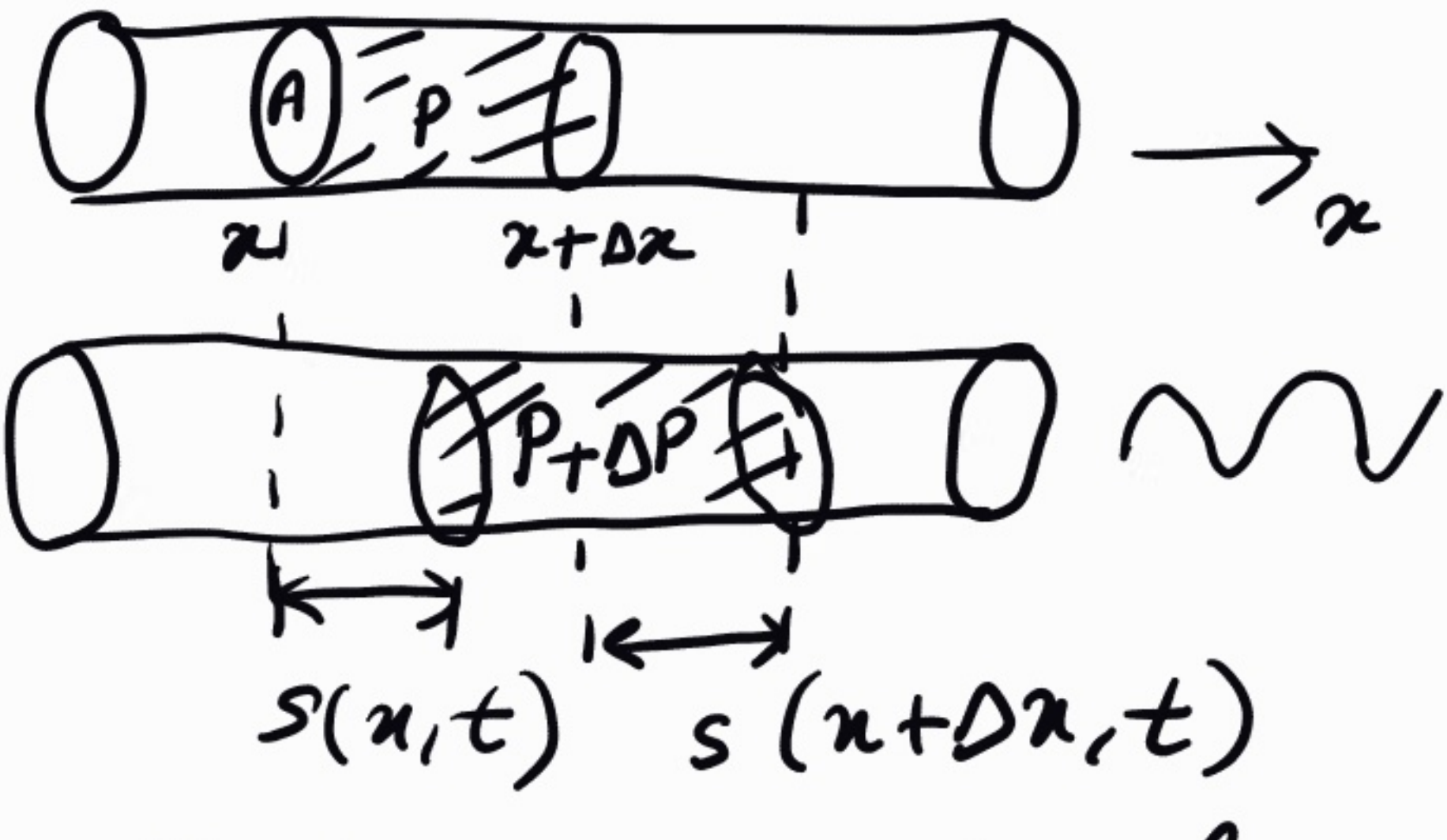
medium	B (Pa)
Hg (20°C)	2.85×10^{10}
Air (0°, 1 atm)	1.485×10^{10}





$$\frac{\Delta P}{\Delta V} \rightarrow \frac{dP}{dV} \Rightarrow B = -V_0 \frac{dP}{dV}$$

Proof : $\Delta p_m = (v \rho \omega) s_m$



- Fluid under additional pressure ($+\Delta P$) → expands.
- Displacement $s(x,t)$ is sinusodial at two ends of the fluid cylinder → different displacements.

Δ in Volume of the cylinder after Pressure increment

$$\Delta V = A \times [S(x+\Delta x, t) - S(x,t)]$$

$$= A \Delta x \frac{\partial S}{\partial x}$$

$$\frac{\partial s}{\partial x} = \lim_{\Delta x \to 0} \frac{S(x+\Delta x) - S(x)}{\Delta x}$$

Initial fluid Volume : $V = A\Delta x$

$$\frac{\Delta V}{V} = \frac{A \cancel{\Delta x} \frac{\partial s}{\partial x}}{\cancel{A \Delta x}} = \frac{\partial S}{\partial x} \quad \text{——— (1)}$$

$$= -\frac{\Delta P}{B} \quad \text{——— (2)}$$

$$\therefore -\Delta P / B = \partial S / \partial x$$

Let $s(x,t) = s_m \cdot \cos(kx - \omega t)$

$$-\frac{\Delta P}{B} = \frac{\partial}{\partial s}\left(s_m \cdot \cos(kx - \omega t)\right)$$

$$= -k s_m \cdot \sin(kx - \omega t)$$

Let $\Delta P = \Delta P_m \sin(kx - \omega t)$

$$\rightarrow \frac{-\Delta P_m \sin(kx - \omega t)}{B} = -k s_m \cdot \sin(kx - \omega t)$$

$$-\Delta P_m = -B k s_m$$

Since $B = \rho v^2$

$$\therefore \Delta P_m = \rho v^2 k s_m \quad \text{(Proof)}$$

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Wave Speed in an Ideal Gas

⇛ v ~ speed of sound in an ideal gas

$$V = \sqrt{\frac{\gamma RT}{M}}$$

$T \rightarrow$ absolute temperature (K)

$R \Rightarrow$ gas constant ~ 8.31 J/mol·K

$M \Rightarrow$ molar mass (kg/mol)

$\gamma \rightarrow$ ratio of heat capacities

[Cp / Cv]

$\gamma \approx 1.67$ (monoatomic gases)

$= 1.40$ (diatomic ")

$= 1.3$ (triatomic ")

Avogadro's number:

$N_A = 6.022 \times 10^{23}$ for 1 mole of substance

Molar mass:

$M = N_A \cdot m$ → mass of a molecule (kg)

Example: Speed of sound on Mars.

→ Typical martian morning $T \approx -100°C$ (atmosphere made entirely of CO_2). Determine:

i) The speed of sound at such a morning.

ii) The approx. T(K) when the speed would double its value.

Solution.

CO_2 has $\gamma = 1.3$ (triatomic)

$T = 273 - 100 = 173\,K$

$$M_{CO_2} = M_C + 2M_O \rightarrow \begin{bmatrix} \text{molar} \\ \text{mass of} \\ O \end{bmatrix}$$

$$\begin{bmatrix} \text{molar mass} \\ \text{of } C \end{bmatrix}$$

$$= 12\ g/mol + 2(16\ g/mol)$$

$$= 44\ g/mol$$

$$v = \sqrt{\frac{\gamma RT}{M_{CO_2}}} \approx 210\ m/s$$

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$R = 8.31\ J/mol.K$

Wave Speed In Solid

→ depends on elastic properties of material.

→ sound travels faster in stiffer materials. (Y)

$$v = \sqrt{\frac{Y}{\rho}}$$

← Young's modulus or elastic modulus

→ material density



$$\varepsilon = \frac{\Delta L}{L}$$

(strain)

$$Y = [\sigma / \varepsilon]$$



$$\sigma = \frac{F}{A}$$

(stress)

[material deformation]

e.g. Al : $Y = 7 \times 10^{10}\ (N/m^2)$